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Ideological Power and Legitimation

Part IV of *The Six Pillars of the Beast System*

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Abstract

Every governance system must settle who shapes the beliefs on which cooperation depends. This paper treats that question as the ideological pillar of a six-pillar model of governance and asks two things of it: whether the pillar is necessary, and when it becomes an instrument of capture. From a coordination axiom, together with membership turnover and the cross-cultural variability of rules, we derive that every persistent collective implements a recognition function: a recurring public practice that makes the operative rules common knowledge. The derivation establishes the necessity of a function, not of any institution, and it does not establish that belief in the rightness of rules is necessary. Stateless and egalitarian societies implement the function through talk, ridicule, ritual and kinship idiom, without a designated office. A Bayesian endorsement model then shows that a captured authority's endorsement carries no information, that independent authorities raise expected public confidence in sound rules and lower it for unsound ones, and that coordinated control of several authorities lowers the probability that an error is disclosed below the level reached by independent control with the same average independence. We compare how academia, religious institutions, media, schools and medical and scientific bodies hold the function in market-capitalist, communist party-state, social-democratic, Islamic, Catholic-monarchical and secular-democratic regimes, and we state hypotheses with falsifiers. The comparative codings are qualitative and drawn from the historical and scholarly record, and each states what a quantitative estimate of concentration and integration would test.

1 Introduction

Every polity has to answer the question of who shapes the beliefs. People cooperate under rules only when they know which rules are in force, which officials hold office, and what others expect of them. Those beliefs do not arise unaided. Someone teaches the young, recites the genealogy, preaches the sermon, accredits the physician, edits the newspaper, peer-reviews the article and announces the verdict. This paper studies that activity as one of six functions that, on the argument of this series, every functioning governance system performs.

The series rests on a single claim: the ideology changes; the institutional functions remain. Six questions recur in every governance system. Who controls the resources (F)? Who makes the laws (P)? Who interprets them (J)? Who shapes the beliefs (I)? Who enforces the rules (C)? Who produces the wealth that sustains everything (W)? The framework adapts the five-function Hierarchy of the Beast System developed by Largo (2026), whose rule function is split here into rule-making and adjudication. The name comes from Revelation 13, which describes a system

binding authority, allegiance and economic participation. In this series the name designates the universal six-function structure itself. It is never a label for any religion, people, profession, regime or person.

Proposition A of the series states that every functioning governance system contains the six functions together with dependency relations among them; Part VII proves it from axioms about collective life. Proposition B states that the functions become concentrated or coordinated so that independent accountability fails: unaccountable concentration U_t and cross-pillar integration $\mathcal{I}_t$ are both high and capture links are documented (Largo, 2026). Proposition A does not imply Proposition B, and Part VII proves the non-implication. The distinction matters for every pillar. The existence of a pillar is a structural fact; its capture is a contingent condition that must be measured.

The ideological pillar poses these tasks in a sharp form. Its institutions include universities, churches, mosques and seminaries, newsrooms, schools, academies of science and medical bodies. These institutions command trust because they are expected to report what they find rather than what rulers prefer. They can check capture by disclosing errors and refusing endorsement. They can also enable capture by lending credibility to rules that serve a narrow coalition. Nothing in this paper implies that expertise, medicine, science or any faith is fraudulent. The source framework says plainly that a hospital is not a propaganda organ and a professor is not an agent of command (Largo, 2026). The analytical question is when the independence that gives these authorities their value is preserved, and when it is lost.

Every persistent collective with membership turnover and locally variable rules implements a recurring public practice that makes its operative rules common knowledge (Theorem 3.5). Entrants do not know the local rules at entry, because rules vary across societies; coordination requires common knowledge of the rules, and private instruction cannot supply it (Lemma 2.5). The derivation establishes the necessity of recognizing the rules. It does not establish the necessity of believing them rightful.

Independent authorities raise public confidence in sound rules and lower it in unsound ones, while a captured authority's endorsement conveys no information (Proposition 6.2). When one coalition controls the appointment of several authorities, an error is less likely to be disclosed than when the same authorities are appointed independently with the same average independence (Proposition 6.6). These results organize a comparison of six regime types and a set of hypotheses, each with a stated falsifier.

The comparison asks how the universal function is distributed, not whether a regime has it. Every regime type has it. This is what makes the comparison even-handed by construction: a Catholic monarchy, an Islamic state, a party-state and a liberal democracy each have teachers, preachers, censors or editors, and accrediting bodies, and each can be asked the same questions about concentration, coupling and contestability.

2 The ideological layer of the multiplex network

2.1 Shared objects

We use the formal objects of the source framework without redefining them (Largo, 2026). The polity at time t is a directed multiplex network $G_t = (V_t, E_t)$ whose nodes are organizations or coherent actor groups. The series has six layers A_t^m, one for each pillar $m \in \mathcal{M} = \{\mathsf{F}, \mathsf{P}, \mathsf{J}, \mathsf{I}, \mathsf{C}, \mathsf{W}\}$, so $M = |\mathcal{M}| = 6$. An edge weight $a_{ij,t}^m \in [0, 1]$ records the documented capacity of actor i to constrain a feasible action of actor j through mechanism m. A many-to-many role map assigns actors to pillars. Each actor holds a control share $s_{i,t}^m$ of decision-relevant control in pillar m, with

$\sum_i s^m_{i,t} = 1$, and pillar concentration is the Herfindahl-Hirschman index $H_{m,t} = \sum_i (s^m_{i,t})^2$.

Concentration is dangerous only where it escapes review, and the source framework measures review separately. Contestability $Q_{m,t} \in [0,1]$ summarizes transparency, appeal rights, independence, competition and credible replacement of decision makers. The illustrative index of unaccountable concentration is $U_t = \sum_m \omega_m H_{m,t}(1 - Q_{m,t})$ with preregistered weights $\omega_m \geq 0$ summing to one.

Integration across pillars is a separate quantity from concentration within them. Cross-pillar integration $C_{mn,t} \in [0,1]$ measures how far one coalition commands appointments, budgets, sanctions or agenda in both m and n, and the integration statistic is

$$\mathcal{I}_t = \frac{2}{M(M-1)} \sum_{m<n} C_{mn,t} = \frac{1}{15} \sum_{m<n} C_{mn,t}. \tag{1}$$

A capture link $(i \to j, m)$ is influence that meets three conditions: privileged access or discretion, a resulting weakening of applicable accountability or contestability, and material evidence tying the asymmetry to an institutional output. In the aggregate state vector $x_t = (F_t, P_t, J_t, I^{cap}_t, C_t, W_t)$ the ideological capacity is written I^{cap}_t. The pillar label is the sans-serif I and the integration statistic is the calligraphic $\mathcal{I}_t$; the two are never interchanged.

For the ideological layer, an edge $a^{\mathsf{I}}_{ij,t}$ records actor i's documented capacity to constrain j by shaping what j or j's audience believes about rules, officials or facts. Accrediting a university, licensing a physician, listing a book on an index of prohibited titles, issuing a juristic opinion on a ruler's decree, withdrawing a broadcast license and retracting an article are examples. Persuasion with no constraining capacity, such as an op-ed that readers may ignore at no cost, is not an edge in this layer.

2.2 Legitimation and legitimating authorities

Let R_t denote the rule system in force in a collective at time t, in the sense of Definition 2.1 of Part II: the general prescriptions that members are expected to follow and that officials, where they exist, apply. The rule system includes rules that designate who holds office and rules that specify sanctions.

Definition 2.1 (Legitimation function)**.** The *legitimation function* of a collective is the class of practices that produce, transmit and validate the beliefs about R_t, about who holds authority under R_t, and about relevant facts, on which members' coordination depends. A practice belongs to the class if it regularly changes members' beliefs about these matters and if members rely on the changed beliefs when deciding whether to comply or cooperate.

The definition names a function, not an institution. A campfire conversation, a liturgy, a curriculum, an evening news broadcast, a peer-reviewed journal and a pharmacopoeia can all belong to the class. Two contents of the function, which the literature often merges, need to be kept apart.

Definition 2.2 (Recognition and normative legitimacy)**.** A belief system supports *recognition* of R_t when members can identify which rules are in force and who is entitled to make, apply and enforce them. Following Hart (1961), the criteria members use for this identification form the collective's *rule of recognition*. A belief system supports *normative legitimacy* when, in addition, members regard the rules or the officials as rightful, so that the reason for compliance includes that judgment and not only the expectation of sanction or the convenience of conformity.

Hart introduced the rule of recognition as the remedy for the uncertainty of a regime of primary rules alone; the rules of change and of adjudication remedy its static character and its inefficiency

(Hart, 1961). In the series, Part II treats the rule of change, the procedure by which a collective alters its rules (Definition 3.6 of Part II), and Part III the rule of adjudication. This Part treats the rule of recognition, which is the minimal content of the ideological function. Normative legitimacy is the richer content that Weber (1978) analyzed in his types of legitimate domination; Part II applies those types to the question of who makes rules.

Definition 2.3 (Legitimating authority)**.** A *legitimating authority* is an actor whose public statements about R_t, about officials or about facts relevant to compliance move other members' beliefs, because members expect those statements to track the truth or the tradition rather than the speaker's interest. We write $\mathcal{K}_t$ for the set of legitimating authorities at time t.

Clergy and ulama, teachers and universities, academies and journals, newsrooms, professional and medical associations, regulatory agencies in their certifying role, and the propaganda departments of ruling parties are legitimating authorities under this definition. The definition does not require that an authority be accurate; it requires that its audience treats it as tracking something other than its own interest. That expectation is the asset that capture spends.

Healthcare and science enter this Part only in their knowledge-producing and standard-setting role. Clinical guidelines, drug approval, accreditation of medical schools and the publication of trial results shape what patients, physicians and governments believe. The delivery of care is production and belongs to Part VI. The cultural authority of medicine, which Starr (1982) traced in the United States, and the professional autonomy that Freidson (1970) analyzed are objects of this layer because they govern whose statements about health count.

2.3 Knowledge, common knowledge and public transmission

Coordination depends on more than each member knowing the rules. Each must know that the others know, and so on. The standard formalism makes the requirement precise (Lewis, 1969; Aumann, 1976).

Definition 2.4 (Knowledge and common knowledge)**.** Let Ω be a set of states and N a finite set of members. Each member $i \in N$ has an information partition $\mathcal{P}_i$ of Ω; $\mathcal{P}_i(\omega)$ is the cell containing ω. For an event $E \subseteq \Omega$, member i *knows* E at ω if $\mathcal{P}_i(\omega) \subseteq E$, and $K_i E = \{\omega : \mathcal{P}_i(\omega) \subseteq E\}$. Write $KE = \bigcap_{i \in N} K_i E$ for mutual knowledge. The event E is *common knowledge* at ω if $\omega \in K^n E$ for every $n \geq 1$, where $K^1 E = KE$ and $K^{n+1} E = K(K^n E)$. An event F is *public* (evident to all) if $F \subseteq K_i F$ for every $i \in N$: whenever F occurs, every member knows that it has occurred. A transmission of R_t is *public* if it generates a public event F with $F \subseteq K_i E$ for every i, where E is the event that R_t is in force; otherwise it is *private*.

Rituals, assemblies and broadcasts that everyone watches together are the familiar sources of public events. Chwe (2001) argued that a central purpose of ritual and public ceremony is to generate common knowledge, and Schelling (1960) showed how shared salience selects among equilibria of coordination games. The following lemma states the logical core.

Lemma 2.5 (Public and private transmission)**.** *Let $E \subseteq \Omega$ be an event.*

(a) If F is a public event and $F \subseteq K_i E$ for every $i \in N$, then E is common knowledge at every $\omega \in F$.

(b) Conversely, if E is common knowledge at ω, then the set F^* of states at which E is common knowledge is a public event containing ω, and $F^* \subseteq K_i E$ for every i.

(c) Mutual knowledge does not imply common knowledge: there are partitions and a state at which every member knows E but some member does not know that another knows E.

Proof. Each operator K_i is monotone: if $A \subseteq B$ then $K_i A \subseteq K_i B$, because a cell contained in A is contained in B.

(a) We show $F \subseteq K^n E$ for all n by induction. For $n = 1$, $F \subseteq K_i E$ for every i gives $F \subseteq KE$. Suppose $F \subseteq K^n E$. For each i, publicity and monotonicity give $F \subseteq K_i F \subseteq K_i(K^n E)$. Intersecting over i yields $F \subseteq K^{n+1} E$.

(b) Let $F^* = \bigcap_{n \geq 1} K^n E$. Since $K^1 E \subseteq K_i E$, $F^* \subseteq K_i E$. Take $\omega' \in F^*$ and any i. For each n, $\omega' \in K^{n+1} E \subseteq K_i(K^n E)$, so $\mathcal{P}_i(\omega') \subseteq K^n E$ for every n, hence $\mathcal{P}_i(\omega') \subseteq F^*$. Thus $\omega' \in K_i F^*$, and F^* is public.

(c) Let $\Omega = \{\omega_1, \omega_2, \omega_3\}$, $N = \{1, 2\}$ and $E = \{\omega_1, \omega_2\}$. Member 1 is told privately whether E holds: $\mathcal{P}_1 = \{\{\omega_1, \omega_2\}, \{\omega_3\}\}$. Member 2 is told only at ω_1: $\mathcal{P}_2 = \{\{\omega_1\}, \{\omega_2, \omega_3\}\}$. At ω_1 both members know E. But $K_2 E = \{\omega_1\}$ and $K_1 K_2 E = \emptyset$, because member 1's cell $\{\omega_1, \omega_2\}$ is not contained in $\{\omega_1\}$. Member 1 knows the rule but does not know that member 2 knows it. □

Part (b) is a form of Aumann's characterization of common knowledge (Aumann, 1976). Together, (a) and (b) say that common knowledge of the rules is available exactly when some public event certifies them. Part (c) shows why private instruction, however thorough, does not suffice. A teacher who instructs each pupil separately produces knowledge; an assembly in which pupils see one another being instructed produces common knowledge.

Remark 2.6. Common knowledge is an idealization. Rubinstein (1992) showed in the electronic mail game that any finite number of levels of mutual knowledge can fail to support coordination that common knowledge would support. Monderer and Samet (1989) showed that an approximate notion, common p-belief, restores continuity: equilibria that require common knowledge are approximated by equilibria under common p-belief for p close to one. The arguments below go through with public events replaced by events that each member believes with probability at least p whenever they occur. We state them for common knowledge to keep the logic visible.

2.4 Concentration and contestability in the ideological layer

The control share $s^{\mathsf{I}}_{i,t}$ is a share of decision-relevant control over authoritative outputs: appointment and removal of authorities, accreditation and licensing, the content of curricula, approval of doctrinal or scientific standards, and the reach of broadcast and print channels. Belief itself is not a fixed quantity to be divided, so s^{I} is defined over these outputs and not over minds. Concentration $H^{\mathsf{I}}_t := H_{\mathsf{I},t}$ is high when one organization appoints the clergy, writes the curriculum, licenses the presses and certifies the professions; it is low when these decisions are spread across organizations with separate appointment and funding.

Definition 2.7 (Components of ideological contestability)**.** The ideological contestability $Q^{\mathsf{I}}_t := Q_{\mathsf{I},t}$ is a preregistered aggregate, with values in $[0, 1]$, of five components: (i) *plurality*: the number and audience reach of legitimating authorities whose appointment and funding are independent of one another; (ii) *independence*: tenure protection, editorial autonomy, academic freedom and juristic or doctrinal autonomy of authorities from the governing coalition; (iii) *disclosure*: publication of the funding, sponsorship and conflicts of interest of authorities; (iv) *reviewability*: procedures by which an authority's statements can be challenged, replicated, corrected or retracted, including peer review, rights of reply and public correction; (v) *exit*: the ability of audiences to turn to rival authorities without legal or material sanction, the ideological case of exit in the sense of Hirschman (1970), leaving an organization rather than voicing complaint within it, which Part VI defines (Definition 2.5 of Part VI).

The components are not new. They restate the source framework's contestability for the ideological layer and match the checks it lists for its knowledge mechanism: independent epidemiology, investigative reporting, protected academic freedom and professional resistance (Largo, 2026). Their value lies in making the aggregate auditable: each component can be coded from documents, and a disagreement about Q^{I} can be traced to a disagreement about one of them.

Ideological concentration alone cannot produce a strong classification under the source criterion, which needs both high U_t and high $\mathcal{I}_t$. The bound is elementary but worth stating. If every pair not involving I has $C_{mn,t} = 0$, then by (1) $\mathcal{I}_t = \frac{1}{15}\sum_{n\neq \mathsf{I}} C_{\mathsf{I}n,t} \leq 5/15 = 1/3$. A polity whose governing coalition controls every belief-producing institution but none of the other five pillars shows ideological fusion, not system-wide integration. Capture of belief becomes system-wide only through coupling to appointments, budgets or sanctions in other pillars, which is why Section 6 studies the couplings.

The value of a single authority to its audience depends on how independent it is of whoever controls it. Section 6 models this dependence with an independence parameter $\lambda \in [0, 1]$, the probability that an authority reports its own findings rather than the position of a controlling coalition. The components of Q^{I} are the institutional means of raising λ.

3 Necessity of the legitimation function

3.1 Assumptions

The series derives each pillar from an axiom about collective life. Part VII owns the axiom set and proves the general necessity theorem. The ideological pillar rests on the coordination axiom. We state it with the auxiliary assumptions that the derivation needs, and we mark which assumptions are empirical.

Fix a collective G with members N_t at time t and rule system R_t. Say that G *functions* at t if members cooperate on the basis of R_t: they comply with its rules and recognize its office holders at least often enough for the collective to make and carry out collective decisions. Let $E_t \subseteq \Omega$ be the event that R_t is the rule system in force at t. Information changes over time, so member i has a partition $\mathcal{P}_{i,s}$ and knowledge operator $K_{i,s}$ at each date s. Write τ_i for i's entry date. For $s < \tau_i$ we set $\mathcal{P}_{i,s} = \{\Omega\}$, the trivial partition: before entry a person knows nothing about the local rules beyond what is true in every state. Members do not forget: for $s \leq s'$, $\mathcal{P}_{i,s'}$ refines $\mathcal{P}_{i,s}$. Common knowledge at date s means common knowledge under the date-s partition profile $(\mathcal{P}_{j,s})_j$.

Assumption 3.1 (Ax4, coordination)**.** If G functions at t, then there is a set $N_t^* \subseteq N_t$ of members, large enough that cooperation among them sustains the collective, such that E_t is common knowledge among N_t^* at the true state at date t.

Assumption 3.2 (Turnover)**.** There is a horizon L such that for every t at least one member of N_{t+L}^* entered the collective during $[t, t + L]$.

Assumption 3.3 (Non-innateness)**.** At entry, an entrant cannot tell which rule system will be in force at any later date: for every entrant e, every date $s \geq \tau_e$ and the true state ω, $\mathcal{P}_{e,\tau_e}(\omega) \not\subseteq E_s$.

The coordination axiom is the series' Ax4 in the precise form that Lewis (1969) gave to conventions: a convention is a regularity sustained because it is common knowledge that others conform. Turnover is universal in human collectives, whose members are born and die, marry in and out, and join and leave; the assumption adds only that some entrants join the cooperating core within a bounded time. Non-innateness is an empirical claim with strong support. Rule systems differ widely across human societies, as every ethnography in Table 1 shows, so an entrant cannot

read the local rules off a genetic or otherwise fixed endowment. The assumption fails for social insects, whose coordination rules are largely fixed by development, and that failure marks the scope of the theorem.

Lemma 3.4 (Renewal)**.** *Under Assumptions 3.2 and 3.3, for every t there is a member $e \in N^*_{t+L}$ with $t \le \tau_e \le t+L$ who does not know E_{t+L} at date τ_e.*

Proof. Assumption 3.2 supplies $e \in N^*_{t+L}$ with $\tau_e \in [t, t+L]$. Since $t+L \ge \tau_e$, Assumption 3.3 gives $\mathcal{P}_{e,\tau_e}(\omega) \not\subseteq E_{t+L}$, that is $\omega \notin K_{e,\tau_e} E_{t+L}$. □

Renewal can also arise without entry. If the rule system changes within the window through the rule of change, the procedure by which a collective alters its rules (Definition 3.6 of Part II), and the change was not announced before the window began, then some member did not know the new rule system at the window's start. The theorem below holds with that case in place of Assumptions 3.2 and 3.3.

3.2 The necessity lemma

Theorem 3.5 (Necessity of recognition)**.** *Let G satisfy Assumptions 3.1, 3.2 and 3.3 and function at every date of an interval longer than L. Then within every window $[t, t+L]$ inside the interval:*

(i) the partition of at least one member $e \in N^*_{t+L}$ is refined between τ_e and $t+L$ by information bearing on the rule system, from a cell not contained in E_{t+L} to a cell contained in it; and

(ii) the partition profile of N^*_{t+L} changes during the window from one under which E_{t+L} is not common knowledge at the true state to one under which it is, so that at date $t+L$ there is an event, public under the date-$(t+L)$ profile, that contains the true state and is contained in $\bigcap_{i \in N^*_{t+L}} K_{i,t+L} E_{t+L}$.

The practices that produce these refinements belong to the legitimation function of Definition 2.1. Hence the legitimation function is implemented in G, at least in the form of recognition.

Proof. Fix a window $[t, t+L]$ inside the interval and let ω be the true state. By Assumption 3.1, E_{t+L} is common knowledge among N^*_{t+L} at ω at date $t+L$. By Lemma 2.5(b), applied to the date-$(t+L)$ profile, the set F^* of states at which E_{t+L} is common knowledge under that profile is public under it, contains ω, and satisfies $F^* \subseteq K_{i,t+L} E_{t+L}$ for every $i \in N^*_{t+L}$.

By Lemma 3.4, some $e \in N^*_{t+L}$ with $\tau_e \in [t, t+L]$ has $\omega \notin K_{e,\tau_e} E_{t+L}$. Since $\omega \in F^* \subseteq K_{e,t+L} E_{t+L}$, and since members do not forget, $\mathcal{P}_{e,t+L}(\omega) \subseteq \mathcal{P}_{e,\tau_e}(\omega)$, e's cell at ω shrank between τ_e and $t+L$ from a set not contained in E_{t+L} to a set contained in it, which is (i).

Under the date-τ_e profile, E_{t+L} is not common knowledge among N^*_{t+L} at ω, because e does not know it and common knowledge implies mutual knowledge. Under the date-$(t+L)$ profile it is, with F^* as the certifying public event. The change lies entirely in the refinement of members' partitions within the window, which is (ii). The practices that produced the refinement regularly change members' beliefs about the rule system, and members rely on the resulting common knowledge to coordinate, which is the defining property in Definition 2.1. Since the argument applies to every window of length L, such practices recur. □

The theorem is modest in its mathematics and specific in its content. Its mathematical core is the characterization of common knowledge by public events. Its empirical content sits in the three assumptions, all of which the proof uses, which is where the series' standard of proof requires it to sit: the necessity claim is derived from stated premises about collective life, and it can fail if

the premises fail. Part VII states the general theorem for all six pillars and the non-implication of Proposition B; the present result is the pillar-level piece for I.

Remark 3.6 (What the theorem does not show). The theorem shows that recognition is necessary. It does not show that normative legitimacy is necessary. A collective in which members know which rules are in force and comply from fear or habit satisfies every assumption above. Comparative research on autocracies treats legitimation as one of three supports of stable rule, alongside repression and co-optation, and finds that the mix varies (Gerschewski, 2013). Wintrobe (1998) modeled dictators who trade off loyalty against repression, and Weber (1978) acknowledged that obedience can rest on fear or interest as well as on belief in legitimacy. Normative legitimacy is a frequent and valuable content of the ideological function, but it is a contingent one. This narrowing matters for the comparison: a regime with low normative legitimacy still has a fully implemented ideological pillar, because it still must make its rules, sanctions and officials common knowledge.

3.3 Minimal forms in stateless and consensus societies

The strongest test of a necessity claim is the set of societies described as having no state, no chiefs with coercive power, or radical equality. The theorem predicts that each implements recognition through some public practice, even with no designated office.

Proposition 3.7 (Minimal form; empirical)**.** *In each of the stateless, egalitarian or consensus-based societies of Table 1, the legitimation function is implemented by public practices without a designated office, and decision-relevant control over those practices is widely dispersed, so that H^{I} is low and Q^{I} is high.*

This is an empirical proposition, supported by the ethnographic record rather than proved. The support is summarized in Table 1 and discussed below.

The Ju/'hoansi are the case most often cited as lacking hierarchy. Their norms of sharing, modesty and restraint are sustained by talk. Wiessner (2005) recorded conversations in Ju/'hoansi camps and found that criticism of norm violations was mostly verbal and directed at behavior such as stinginess and self-aggrandizement. Talk in a small camp has the structure of a public event: speakers and listeners are co-present, and each sees who else hears. Lee (1979) and Lee (2013) describe the custom of deprecating a successful hunter's kill, which teaches the hunter and everyone listening that boasting will not be tolerated. No one holds an office of teacher or censor. The function is distributed across the whole adult population, so H^{I} is close to its minimum and exit to another camp keeps Q^{I} high.

The Nuer were described by Evans-Pritchard (1940) as an ordered anarchy without government. The ideological function is present in the lineage idiom that tells every Nuer how segments relate and which obligations follow, and in ritual specialists such as the leopard-skin chief, whose standing derived from a ritual relation to the earth rather than from any power to command. The case confirms that the function can exist without an office that issues orders.

Clastres (1987) described Amazonian chiefs whose authority rested on generosity and speech. The chief's daily discourse recalls the traditional norms of the group; it is a public recitation, not a command, and a chief who tries to command loses his following. Clastres read these societies as actively preventing the political function from separating from society. The reading is consistent with Theorem 3.5: the function exists, its holder has no power to enforce it, and concentration is prevented by design. Clastres' evidence has been criticized as selective, and the case should be read with that caution.

Boehm (1999) argued that mobile foragers maintain equality through a reverse dominance hierarchy in which the group as a whole suppresses would-be dominators by gossip, ridicule,

Table 1: Minimal forms of the legitimation function in stateless and egalitarian societies. Each row names the public practices that make rules and expectations common knowledge and the strongest reading that the function is absent.

Case	Public practices that carry recognition	Strongest counter-reading and reply
Ju/'hoansi	Camp talk, joking and criticism heard by all, the leveling custom of insulting the meat, and the healing dance (Lee, 1979, 2013; Marshall, 1976; Wiessner, 2005)	No office holds authority. Reply: the function is carried by talk that everyone hears, which is exactly a public event.
Nuer	Lineage idiom shared across segments, ritual specialists whose standing is religious rather than political, and prophets (Evans-Pritchard, 1940)	Evans-Pritchard stressed the absence of government. Reply: the function is present while the office is absent.
Amazonian chiefdoms	The chief's duty of public speech, which recalls traditional norms, and myth shared by the group (Clastres, 1987)	The chief commands nothing. Reply: the chief's speech is a public recitation of norms without power to command.
Mobile foragers	Moral talk and gossip that circulate shared norms and identify would-be dominators (Boehm, 1999, 2012)	Equality might seem to require no shared ideology. Reply: the egalitarian ethos is itself a shared and enforced belief.
Immediate-return societies	Public leveling practices such as obligatory sharing and mockery of self-promotion (Woodburn, 1982)	Ritual is egalitarian and unspecialized. Reply: unspecialized is not absent.
Commons users	Rules known to appropriators, collective-choice meetings and public monitoring (Ostrom, 1990)	Users may coordinate by self-interest alone. Reply: Ostrom's design principles include rules that users know and can change.

ostracism and, in extreme cases, killing. The sanctions belong to Part V. The shared moral understanding that identifies what counts as domination belongs here. Woodburn (1982) made the parallel point for immediate-return societies: equality there is an institutional outcome produced by leveling practices, not the absence of institutions.

Ostrom (1990) showed that long-enduring commons are governed by rules that users know, can modify and monitor. The design principles are institutional, not ideological, but each presupposes that the operative rules are common knowledge among appropriators. Graeber and Wengrow (2021) added that some societies alternated seasonally between hierarchical and egalitarian arrangements, each publicly marked. Their claims are contested and are cited here only for the narrow point that political arrangements were publicly recognized as such.

These cases come with standard caveats. Contact-era ethnography observed groups in marginal environments, sometimes idealized them, and generalized from few cases (Kelly, 1995). None of these caveats affects the narrow claim of Proposition 3.7. Each caveat would matter if the claim were that these societies show the original condition of humanity; the claim is only that they show the function without the office.

3.4 Counterexample attempts

The most serious attempt to defeat the necessity claim is coordination by force alone. On this view a sufficiently strong coercive apparatus needs no shared beliefs; it needs soldiers and prisons. The reply is that force itself depends on common knowledge. A sanction rule deters only if the people it targets know it, and a coercive organization holds together only if its members expect one another to carry out orders. Kuran (1991) showed that regimes resting on public displays of support can

collapse suddenly when private opposition becomes common knowledge, which is evidence that rulers depend on controlling what is common knowledge even when they rely on force. Svolik (2012) treats the control of information among elites as a central problem of authoritarian rule. The attempt fails for recognition. It succeeds against a stronger thesis, that normative legitimacy is necessary, which Remark 3.6 has already abandoned.

A second attempt holds that markets coordinate through prices without shared beliefs. Prices do economize on information. But exchange presupposes common knowledge of property conventions: who owns what, what counts as payment, and which contracts are binding. Those conventions are part of R_t, and they are taught, displayed and defended. Hayek (1973) himself stressed that the order of a market society rests on rules of just conduct that people follow without being able to state them; following a rule one cannot state is compatible with common knowledge of the practice it supports.

A third attempt notes that in a small face-to-face group everyone observes everyone, so common knowledge arises from co-presence without any practice designed to produce it. This is correct and is the minimal form, not a counterexample. Co-present talk is a practice that produces public events.

A fourth attempt points to coordination without beliefs in animal societies. The non-innateness assumption excludes it, and the exclusion is principled: where rules are fixed by development, no transmission is needed. The theorem is a claim about human governance, where rules vary.

The falsifier follows from the assumptions. The necessity claim would be refuted by a persistent human collective, with turnover and with rules that differ from its neighbors', in which entrants come to coordinate on the local rules without any practice that informs them of the rules and without any public event that certifies them. We know of no such case in the sources reviewed.

3.5 Function and institution

Two classic errors of functionalism must be avoided. Hempel (1959) showed that from the premise that a function is needed one cannot infer that a particular institution exists, because functional equivalents may meet the need. Theorem 3.5 respects this limit: it asserts that some practice in a class is implemented, and the class includes campfire talk and state broadcasting alike. Elster (1983) argued that explaining an institution by its beneficial effects requires a feedback mechanism from effect to institution. The theorem is not such an explanation. It is a condition of persistence: collectives that lack the function do not function, whatever the reason the function arose. It does not explain why a given regime has a ministry of religious affairs or a public broadcaster, and the comparative sections never use it for that purpose. Merton (1968) supplies the remaining vocabulary: the manifest function of a university is teaching and research, and its latent function in the ideological layer may include certifying elites, which is a separate empirical question.

4 Literature on ideological power

4.1 Consent and coercion

The modern study of ideological power begins with the observation that rule rests on consent as well as force. Gramsci (1971) called the organization of consent through civil society hegemony and argued that in societies with dense churches, schools, associations and presses, rulers depend on these institutions as much as on the state's coercive apparatus. Althusser (1971) sharpened the contrast into a distinction between repressive and ideological state apparatuses, the latter including churches, schools, families, the media and cultural institutions, and he argued that in mature capitalism

the school had displaced the church as the dominant ideological apparatus. Both accounts treat ideological institutions as functional supports of a ruling class. That functional reading is precisely what Elster (1983) criticized, and it is what the present paper avoids: we claim the necessity of a function, not that any given apparatus exists because it serves a class.

Weber (1978) placed belief at the center of domination by distinguishing traditional, charismatic and rational-legal grounds of legitimacy. Part II uses the Weberian types to analyze who makes rules. This Part takes from Weber only the distinction between compliance grounded in belief and compliance grounded in fear or interest, which becomes the distinction between normative legitimacy and recognition in Definition 2.2.

4.2 Faces of power and networks of meaning

Bachrach and Baratz (1962) argued that power is exercised not only in decisions but also in non-decisions, by keeping issues off the agenda. Lukes (2005) added a third dimension: the power to shape preferences so that conflicts of interest never surface. Part II owns agenda control, the power to determine which proposals reach a decision, as an object of rule-making (Definition 2.9 of Part II). The third dimension belongs here, because shaping preferences is shaping beliefs. The difficulty Lukes acknowledged is measurement: claims about preferences that people would hold but for ideological influence require a counterfactual that is hard to observe. The endorsement model of Section 6 sidesteps the problem by studying a narrower and observable quantity, the information content of an authority's statements.

Mann (1986, 1993) gave the most developed comparative-historical account. He distinguished ideological, economic, military and political networks of power and treated none as primary. Ideological power, in his account, derives from the human need for meaning, norms and aesthetic and ritual practice, and it can be transcendent, organizing a community across political boundaries as the world religions did, or immanent, reinforcing the morale of an existing group. Mann's framework is the closest predecessor of the six-pillar model. It lacks a separate judicial pillar and a separate pillar for the producing population, and it has no measure of capture; Part VII compares the frameworks in detail. Its value here is the distinction between transcendent and immanent ideological power, which maps onto the difference between authorities whose appointment and revenue lie outside the polity, such as a transnational church, and authorities created by the polity itself.

Bourdieu (1989, 1991) described symbolic power as the power to constitute the given by stating it, effective to the degree that it is misrecognized as something other than power. His account of the convertibility of economic, cultural and symbolic capital anticipates the cross-pillar coupling C_{FI}, which Part I treats as its pair. Bourdieu and Passeron (1977) applied the account to schooling, arguing that educational systems reproduce class structure while appearing meritocratic.

4.3 The public sphere and the media

Habermas (1989) traced the rise of a bourgeois public sphere of rational-critical debate in eighteenth-century Europe and its later refeudalization, as mass media and organized interests turned a reasoning public into an audience. Fraser (1990) argued that the bourgeois public sphere excluded women, workers and others, who formed subaltern counterpublics of their own. Both accounts bear on the plurality component of Q^{I}: a public sphere is contestable to the degree that counterpublics can form and be heard.

The political economy of the media divides on the question of supply and demand. Herman and Chomsky (1988) proposed a propaganda model in which ownership, advertising, reliance on official sources, organized flak and a dominant ideology filter news in favor of elite interests. The model

predicts that owner and advertiser interests shape content. Gentzkow and Shapiro (2006) showed instead that slant can arise from demand: outlets that wish to appear accurate tilt reports toward their audience's prior beliefs, so slant can exist without any owner's design. Djankov et al. (2003) assembled cross-country data on media ownership and found that the largest media firms were typically controlled by governments or by private families, and that state ownership was associated with weaker press freedom and political rights. Besley and Prat (2006) modeled media capture by government and showed that capture is harder when there are more independent outlets and when bribing them is costly, which links plurality directly to accountability. Hallin and Mancini (2004) classified Western media systems into liberal, democratic corporatist and polarized pluralist models, differing in the strength of commercial media, the degree of political parallelism, journalistic professionalization and the role of the state.

Persuasion effects are measurable. DellaVigna and Kaplan (2007) exploited the staggered entry of a cable news channel into United States towns and found an effect on presidential vote shares. Enikolopov et al. (2011) found that access to the one national television channel independent of the government in Russia's 1999 parliamentary election reduced votes for the government's party and raised votes for opposition parties. These studies establish that the ideological layer has real edges: control of a channel changes political behavior.

Authoritarian information control has its own literature. King et al. (2013) analyzed censorship of Chinese social media and found that posts criticizing the state were often left standing while posts with potential for collective action were removed. Stockmann (2013) found that commercialized Chinese media were perceived as more credible than official outlets and could therefore carry the party's messages more persuasively. Guriev and Treisman (2019, 2022) described a shift from rule by fear to rule by spin, in which leaders manipulate information while simulating democratic forms; Part II treats this as a strategy of rule-makers. Kuran (1991) analyzed preference falsification, in which citizens publicly support regimes they privately oppose, and explained why the revolutions of 1989 surprised observers.

Digital platforms add a layer of intermediaries between authorities and audiences. Gillespie (2019) argued that moderation of what users may post is not incidental to platforms but constitutive of them, so platform firms make continual decisions about which speech reaches whom. Allcott and Gentzkow (2017) found that fabricated news stories circulated widely on social media before the 2016 United States election, while estimated exposure per adult was modest. Guess et al. (2023) replaced the algorithmic feeds of consenting Facebook and Instagram users with reverse-chronological feeds for three months during the 2020 campaign and found large changes in what users saw but no detectable change in polarization or political attitudes. Roberts (2018) showed that Chinese censorship works less through fear than through friction and flooding, which make unwanted information costly to find rather than impossible. Platforms are thus channels whose control shares and contestability can be coded like those of broadcasters, with the difference that ranking and moderation decisions are made at very large scale by few organizations.

4.4 Professions and expert authority

Professions convert knowledge into authority. Larson (1977) described the professional project as an effort to secure market closure and social standing through control of training and credentials. Abbott (1988) treated professions as competitors for jurisdiction over tasks. Freidson (1970) argued that medicine's autonomy, its control over the content of its own work, made it the model profession, and Starr (1982) traced how American medicine converted cultural authority into economic power and professional sovereignty. These accounts explain why expert authority is valuable to rulers: experts certify that a rule is sound in terms the public trusts.

Evidence on sponsorship qualifies that trust without destroying it. Bekelman et al. (2003) reviewed studies of financial conflicts of interest in biomedical research and found that industry sponsorship was associated with conclusions favorable to sponsors. Lexchin et al. (2003) reached a similar conclusion for pharmaceutical sponsorship, and the Cochrane review by Lundh et al. (2017) found that industry-sponsored drug and device studies more often reported favorable results and conclusions than studies with other sponsors. Oreskes and Conway (2010) documented campaigns in which a small number of scientists, with industry support, manufactured doubt about the harms of tobacco and about environmental science. Against these findings stand accounts of why science remains trustworthy. Oreskes (2019) located the reliability of science in its social process of critical vetting by a diverse community, and Carpenter (2010) showed that the United States Food and Drug Administration built autonomy on its reputation for scientific judgment. Slaughter and Rhoades (2004) documented the growing market orientation of universities, which raises the coupling between the financial and ideological pillars. Read together, this literature supports the model of Definition 6.1: expert authorities are informative to the degree that their judgments are independent of their sponsors, and institutions such as disclosure, review and reputation exist to secure that independence.

4.5 Religion and the state

The relation between religious authority and political rule is the oldest problem of the ideological pillar. Stepan (2000) proposed the twin tolerations as the condition for democracy: elected institutions free from religious veto, and religious citizens free to worship and to advance their views in public. Casanova (1994) documented the deprivatization of religion, its return to public life in several countries from the 1980s. Philpott (2007) argued that religious actors' political behavior depends on their institutional independence from the state and on their political theology. Fox (2008) coded religion-state arrangements worldwide. Kuru (2009) distinguished passive secularism, which allows public religious expression, from assertive secularism, which excludes it.

Two debates concern Islamic and Catholic polities directly. Kuru (2019) argued that an alliance between ulama and the state, formed in the eleventh century, helps explain authoritarianism and underdevelopment in many Muslim-majority countries. Rubin (2017) argued that rulers who depended on religious authorities for legitimacy faced different incentives from rulers who depended on economic elites, and that the difference shaped economic divergence. Both arguments are contested, and the series cites them as positions, not settled findings. For Catholic Europe, Grzymala-Busse (2023) argued that the medieval Church served as a template and rival for state building, and Grzymala-Busse (2015) showed that churches whose moral authority is fused with national identity can shape policy through institutional access rather than partisan politics.

4.6 Relation to prior work

Coordination theory (Lewis, 1969; Schelling, 1960; Aumann, 1976; Chwe, 2001) explains why some practice must make rules common knowledge. The political economy of media and expertise (Besley and Prat, 2006; Gentzkow and Shapiro, 2006; Carpenter, 2010) explains why the value of a statement depends on the independence of its source. Comparative work on religion, media and education describes how the institutions vary. No existing framework combines the three: none derives the function from an axiom, states when its institutions check or enable capture, and compares regimes on the same measures. The source framework provides the measures (Largo, 2026). The present paper adds the derivation and the pillar-level model and applies both to six regime types.

5 Comparative analysis across six regime types

Every regime type implements the ideological function; they differ in which institutions hold it, how tightly one coalition controls their appointments and revenue, and which other pillars that coalition also commands. The comparison below codes these differences qualitatively from secondary sources. No numerical value of H^{I}, Q^{I}, U_t or $\mathcal{I}_t$ is reported for any polity, because no preregistered coding exists.

The couplings of the ideological pillar to finance, rule-making and adjudication (C_{FI}, C_{PI}, C_{JI}) are analyzed primarily in Parts I, II and III; this paper treats the couplings to coercion and production (C_{IC}, C_{IW}).

5.1 Market-capitalist societies

In market-capitalist societies the ideological function is held by many organizations with separate ownership and funding: privately owned media firms financed by advertising and subscriptions, public and private universities, foundations and think tanks, professional associations, scientific journals, regulatory agencies in their certifying role, and religious bodies organized as voluntary associations. Control over authoritative outputs is spread across these organizations, so H^{I} is moderate overall, but it varies by sector. Ownership of the largest media firms is often concentrated in a few families (Djankov et al., 2003), while accreditation of physicians or approval of drugs may rest with a single professional body or agency.

The distinctive coupling is between the financial and ideological pillars. Advertising finances news, endowments and grants finance universities, and industry finances a large share of clinical research. Herman and Chomsky (1988) built their propaganda model on this coupling. Slaughter and Rhoades (2004) documented the movement of universities toward market revenue, and Starr (1982) described how medical authority was converted into economic power and later confronted corporate organization of care. Part I treats C_{FI} as its pair and measures the conversion of financial control into favorable outputs of other pillars by a conversion elasticity (Definition 6.1 of Part I); here the question is what the coupling does to the independence parameter λ of Definition 6.1.

The documented capture episodes are sectoral. Systematic reviews associate industry sponsorship with conclusions favorable to sponsors (Bekelman et al., 2003; Lexchin et al., 2003; Lundh et al., 2017). Oreskes and Conway (2010) documented organized efforts to manufacture doubt about tobacco harms and environmental hazards. Each case meets the three conditions of a capture link: privileged access through funding, weakened contestability through selective publication or orchestrated doubt, and material evidence linking the asymmetry to an output, whether a published conclusion or a delayed regulation.

Platform firms now hold part of the function in this regime type. A few companies decide, through ranking and moderation, which statements by journalists, scientists and clergy reach large audiences (Gillespie, 2019). In the terms of Section 2, they hold a large share of control over reach, one of the outputs that defines s^{I}, so they raise measured concentration even when the number of speakers grows. The evidence on their persuasive effect is mixed: fabricated stories spread widely (Allcott and Gentzkow, 2017), but an experimental change of feed ranking during an election left attitudes unchanged (Guess et al., 2023). Control over reach is established; its translation into belief is an open empirical question.

The documented checks are also sectoral. Competition among outlets makes media capture costly (Besley and Prat, 2006). Demand-side slant (Gentzkow and Shapiro, 2006) is a rival explanation for partisan coverage that does not require owner control, so evidence of slant is not by itself evidence of capture. Disclosure rules, peer review, replication and the reputational autonomy of

regulators (Carpenter, 2010) raise λ. The capture episodes above were themselves discovered through litigation, journalism and meta-research, which is what source hypothesis H4 predicts: higher rates of documented error discovery can signal stronger accountability rather than more capture (Largo, 2026).

5.2 Communist party-states

In communist party-states the ruling party claims the ideological function as part of its leading role. Party propaganda departments, state newspapers and broadcasters, party schools, academies of sciences, writers' and artists' unions and a compulsory curriculum in Marxism-Leninism hold the function. Kenez (1985) described how the Bolsheviks built a propaganda state in their first decade, combining agitation, literacy campaigns and new media under party direction. Brooks (2000) showed how the Soviet press under Stalin organized public culture as a ritual in which achievements were presented as gifts from the leadership and citizens were expected to express gratitude.

Concentration is high by design. Appointments of editors, rectors and academy presidents pass through the party's appointment channel, the map from a coalition to the offices it fills in each pillar (Definition 2.8 of Part II), so C_{PI} is high and H^{I} approaches its maximum in the formal structure. Coupling to the coercive pillar is strong: censorship offices and security services enforce the boundaries of permissible expression, and Part V treats the security services themselves. Althusser (1971)'s concept of ideological state apparatuses, developed within Marxism, describes this configuration more literally than it describes the societies Althusser had in mind.

Formal structure overstates actual control. Lane and Ross (1995) found commonalities and divisions within the Soviet ruling elite in its last decade, and divisions within the elite reduce the effective concentration of control over authoritative outputs. Kuran (1991) explained the sudden collapse of East European regimes in 1989 as the release of privately held opposition that public displays had concealed. In the People's Republic of China, King et al. (2013) found that censors removed posts with potential for collective action more consistently than posts criticizing officials, which shows a discriminating rather than total control of expression. Stockmann (2013) found that commercialized media, perceived as more credible than official outlets, could carry party messages more effectively. That finding is an instance of Proposition 6.2: perceived independence raises the information value of a statement, so a ruling party can gain by permitting outlets that appear independent while retaining control over their limits. Guriev and Treisman (2019) described the same logic for informational autocracies in general. Platforms extend the party's reach rather than escaping it: domestic platform firms operate under party supervision, and Roberts (2018) found that censorship relies mainly on friction and flooding, so that most users never meet the information that determined users can still find.

Public benefits are part of the record. The early Soviet literacy campaigns that Kenez (1985) described expanded the population that could read, including what the party did not print. Source Proposition 4, that public benefit is compatible with hierarchy, applies here as elsewhere (Largo, 2026). The documented checks are weak in formal terms: there is no lawful independent press and no protected academic freedom. They consist of elite factions, professional norms within the sciences, and the preference falsification that makes displayed support an unreliable guide to private belief.

5.3 Democratic socialism and social democracy

In social democracies the ideological function is held by a mix of public and private institutions in which public institutions have a large role. Public service broadcasters, public universities,

state-funded schools, the press of political parties and labor movements, and, in several Nordic countries, established Lutheran churches are the main holders. Hallin and Mancini (2004) placed the media systems of Northern and Central Europe in their democratic corporatist model, in which mass-circulation commercial media coexisted with media tied to organized social and political groups, journalism was highly professionalized, and the state intervened actively through press subsidies and public broadcasting while legal norms limited its control over content.

Concentration in broadcasting can be high, because a public broadcaster may hold a large audience share. Contestability then depends on the governance of that broadcaster: who appoints its board, how its budget is set, and whether editors are protected. Coupling to the political pillar runs through these appointment and budget channels, and Part II treats C_{PI} as its pair. Coupling to the productive pillar is distinctive in this regime type. Unions and labor parties historically ran newspapers, educational associations and adult education, so the beliefs of the producing population were shaped partly by organizations that the producers themselves controlled. Berman (2006) argued that social democracy's defining idea was the primacy of politics over markets, an ideological commitment advanced through party organization. Esping-Andersen (1990) argued that universal welfare programs build cross-class support for the welfare state, and Rothstein (1996) analyzed how the Swedish model depended on the administrative implementation of reforms. Universal programs thus have an ideological consequence: they produce shared beliefs about entitlement that sustain the programs themselves.

The capture risk in this regime type is that a long-governing party turns public institutions into party instruments. The documented checks are structural. Hallin and Mancini describe the professional autonomy of journalists and the coexistence of public and commercial media, and the boards of public broadcasters in several countries are composed to represent multiple parties and social groups (Hallin and Mancini, 2004). Established churches in these states coexist with religious freedom, and Fox (2008) shows that state support of religion is common among democracies and does not by itself indicate control of belief.

5.4 Islamic states

Islamic states form a diverse class: the early and Abbasid caliphates, the Ottoman empire, Safavid and modern Iran, and Saudi Arabia differ in doctrine, in the organization of religious scholars and in the relation of scholars to rulers. The common feature is that political legitimacy is derived in part from religious law, which makes the ulama, the scholars of that law, central holders of the ideological function. Madrasas, mosques and their sermons, juristic opinions, and in modern states ministries of religious affairs and state seminaries are the main institutions.

The history of the class contains a separation of religious authority from rule that is directly relevant to independence. Crone and Hinds (1986) argued that early caliphs claimed religious authority as deputies of God and that this authority passed to the scholars over the first centuries of Islam; the failure of the caliph's inquisition, the *mihna*, to impose a doctrine on the scholars marked the transition. The case is a documented check: scholars refused an endorsement that the ruler demanded, and the ruler's successors abandoned the demand. Makdisi (1981) traced the madrasa as an endowed college of law, financed by charitable endowments that gave scholars a revenue base distinct from the ruler's treasury. Berkey (1992) showed that in medieval Cairo the transmission of knowledge remained personal, from teacher to student, even as ruling elites endowed madrasas, which coupled patronage to scholarship without placing scholarly authorization under the patrons' control. Crone (2004) surveyed six centuries of medieval Islamic political thought, which itself records a range of views on the obligations of rulers to scholars and law.

Later states increased the coupling. In the Ottoman empire the learned hierarchy was organized

as a career of salaried officials under the chief jurist, the Şeyhülislam, and Imber (1997) showed how the chief jurist Ebu's-su'ud harmonized sultanic law with the sharia; C_{PI} was correspondingly high, while Barkey (2008) emphasized the empire's reliance on brokered intermediaries and its accommodation of diversity. In Saudi Arabia, Commins (2006) and Al-Rasheed (2002) describe an alliance between the ruling dynasty and the Wahhabi religious establishment, in which scholars legitimated the dynasty and received authority over religious education and public morals. In Iran, Arjomand (1988) analyzed how the revolution of 1979 created a clerical state, building on a doctrine of guardianship by the jurist that Khomeini (1981) had set out, and Menashri (1992) traced the relation between education and state ideology in modern Iran. Here the fusion of religious and political authority raises C_{PI} and C_{JI} toward their maximum, although the Shia clergy's older tradition of plural, independently followed authorities remains a source of internal diversity.

The medieval checks on rulers were juristic pluralism across schools of law, endowed revenue, and scholarly reputation built on teacher-to-student authorization. Gellner (1981) stressed a further division, between urban scripturalist scholarship and rural forms of religious authority, which kept any single body from monopolizing religious legitimacy.

In the modern period the main check has been the fragmentation of religious authority. Eickelman (1992) and Eickelman and Piscatori (1996) showed that mass higher education and new media multiplied the people able to claim religious authority, and Zaman (2002) showed that the ulama adapted to this competition and remained influential.

Whether alliances between ulama and states caused authoritarianism (Kuru, 2019), or whether legitimacy bargains between rulers and religious authorities shaped economic paths (Rubin, 2017), remains an open debate. Across the class, concentration and coupling range from low, where scholars held endowed and plural authority, to very high, where religious authority was absorbed into the state. The class is defined by its source of legitimacy, not by a single configuration.

5.5 Catholic monarchies

In the Catholic monarchies of early modern Europe, Habsburg Spain, Bourbon France and the Papal States, the ideological function was shared between the Church and the crown. The Church supplied doctrine, clergy, sacraments, universities and religious orders; the crown supplied patronage, appointment rights and the coercive backing of religious courts. Hsia (1998) described the Catholic renewal after the Council of Trent, including the expansion of Jesuit colleges, which made the Society of Jesus one of the largest educational networks of the period. Fragnito (2001) and her collaborators documented the work of the Roman congregations that maintained the Index of Prohibited Books and licensed printing in Italy. Eisenstein (1979) argued that print transformed the conditions of religious and scholarly authority, which made licensing and indexing central instruments of control.

The configuration varied. In the Papal States temporal and spiritual authority were held by the same sovereign, so C_{PI} reached its maximum. In Habsburg Spain the Inquisition operated under royal as well as papal authority and coupled the ideological pillar to the judicial and coercive pillars. Kamen (1997) argued that the Inquisition's scale and severity were smaller than the black legend held and that it served local and royal interests as much as doctrinal ones; that revision is itself debated, and the quantitative studies of trial records collected by Henningsen and Tedeschi (1986) supply the evidence on which the debate turns. In Bourbon France, Burke (1992) showed how the image of Louis XIV was manufactured by artists, writers and academies working for the crown, and Bossuet (1990) supplied the scriptural theory of divine-right monarchy. Kantorowicz (1957) traced the older political theology of the king's two bodies on which such theories drew.

The transnational Church was also a check on crowns, because its appointment and revenue lay partly outside any single kingdom. In Mann's terms its ideological power was transcendent

(Mann, 1986), and Grzymala-Busse (2023) argued that rivalry between the papacy and rulers shaped European state building. Within France, Van Kley (1996) traced how conflicts between Jansenists, the Gallican tradition, the papacy and the crown, with the parlements often siding with the Jansenists, eroded the sacral authority of the monarchy before 1789. These were counter-publics in Fraser's sense (Fraser, 1990), operating inside a confessional state.

5.6 Secular democracies

In secular democracies the ideological function is distributed among a legally free press, universities with academic freedom, scientific academies and journals, professional bodies, civil society organizations and religious communities that are either separated from the state or established on plural terms. The model is the public sphere that Habermas (1989) reconstructed, with its later commercialization, and the counterpublics that Fraser (1990) added. Constitutional speech protections and independent courts, analyzed in Part III, set the legal boundary of the layer.

Concentration is low in law and variable in fact. Media markets can be concentrated, and the liberal model of Hallin and Mancini (2004) relies on commercial media with weak public broadcasting. Coupling to the financial pillar runs through advertising and research funding, as in the market-capitalist case. Coupling to the political pillar runs through appointments to public science agencies, school curricula and the governance of public broadcasting. The relation to religion differs across democracies. Kuru (2009) contrasted the passive secularism of the United States, which permits public religious expression, with the assertive secularism of France and Turkey, where the state excludes religion from public institutions; in Turkey the state's directorate of religious affairs administers mosques and appoints their imams, which shows that state control of religious authority is compatible with a secular constitution. Grzymala-Busse (2015) found that churches whose moral authority is fused with national identity, as in Ireland and Poland, influenced policy through institutional access, while Casanova (1994) showed that religions entered public debate as participants rather than as state authorities.

Platform intermediaries raise the same question in secular democracies as in market economies, with one addition: their moderation decisions are taken by private firms under speech protections that bind governments but not, in most jurisdictions, the firms themselves (Gillespie, 2019). Contestability over reach therefore depends on competition among platforms and on their own appeal procedures more than on courts.

The capture risk in secular democracies is less fusion than conversion: financial power converted into credibility through sponsorship, or political power converted into control of a public broadcaster. The documented checks are the twin tolerations (Stepan, 2000), judicial protection of expression, the plurality of outlets that makes media capture costly (Besley and Prat, 2006), and the social process of critical scrutiny that Oreskes (2019) identified in science.

5.7 Summary across regimes

Table 2 summarizes the comparison. Each regime type implements the ideological function, as Theorem 3.5 requires. The types differ in which institutions hold the function, how concentrated control over authoritative outputs is, which other pillars it is coupled to, and what keeps authorities independent.

Table 2: The ideological function across six regime types. Entries are qualitative codings from the sources cited in the text.

Regime type	Main holders	Concentration and coupling	Main sources of contestability
Market capitalism	Private media, universities, journals, professions and certifying agencies	Moderate concentration that varies by sector, with strong financial coupling	Competition among outlets, disclosure, peer review and regulator reputation
Communist party-state	Party propaganda organs, state media, party schools and academies	High formal concentration, with strong political and coercive coupling	Elite factions, professional norms and hidden private opinion
Social democracy	Public broadcasters, public universities, labor and party press, established churches	Concentration high in some channels, with strong productive coupling through labor organizations	Plural boards, protected journalism and coexisting commercial media
Islamic states	Ulama, madrasas, mosques, chief jurists and religious ministries	Ranges from low under endowed plural scholarship to very high under clerical rule	Juristic pluralism, endowed revenue and fragmentation of authority
Catholic monarchies	Church hierarchy, orders, universities, Inquisition, Index and court culture	High crown and Church coupling, with full fusion in the Papal States	Partial independence of the transnational Church and theological conflict
Secular democracies	Free press, universities, academies, civil society and religious bodies	Low legal concentration, with moderate financial and political coupling	Speech protections, plurality of outlets and scientific scrutiny

6 Mechanisms, failure modes and checks

6.1 From coordination to capture

Coordination in the ideological pillar becomes capture when a coalition gains control over what authorities say and that control weakens the review that would otherwise expose it. The source framework's capture link supplies the test (Largo, 2026). In the ideological layer its three conditions take a specific form. Privileged access means that a coalition funds, appoints, licenses or can dismiss an authority, or controls the channel through which it speaks. Weakened accountability means that the access is used to remove or evade review: findings are withheld, dissenting scholars are dismissed, sponsorship is concealed, presses are licensed selectively, or rival authorities are suppressed. Material evidence means documents, such as internal correspondence, censorship directives, appointment records or trial registries, that tie the access to a specific output, such as a published conclusion, a juristic opinion, a curriculum change or a broadcast.

The same test says what is not capture. A church whose doctrine agrees with a government's policy is not thereby captured. Industry-funded research that is disclosed, preregistered and reproduced is not captured. A ruler's endowment of a college is patronage, which becomes capture only if it is used to control what the college certifies. A ministry of education is part of the differentiated structure that Proposition A predicts, and source Proposition 1 holds that differentiation is not integration (Largo, 2026). Public goods are a rival explanation for many configurations of the pillar: a public broadcaster or a state seminary may exist because it supplies information or instruction that markets or voluntary bodies would undersupply.

6.2 Endorsement and independence

The value of a legitimating authority to its audience depends on whether its statements track something other than the interest of whoever controls it. A simple model captures this.

Definition 6.1 (Endorsement model)**.** A rule or policy is *sound* ($\theta = G$) or *unsound* ($\theta = B$), with public prior $\pi = \Pr(\theta = G) \in (0, 1)$. An authority observes a private signal $s \in \{0, 1\}$ with $\Pr(s = 1 \mid G) = \Pr(s = 0 \mid B) = q \in (\frac{1}{2}, 1)$. With probability $\lambda \in [0, 1]$ the authority is *independent* and endorses ($e = 1$) if and only if $s = 1$. With probability $1 - \lambda$ it is *captured* and endorses regardless of s. The public observes e but neither s nor the authority's type, knows (π, q, λ), and forms the posterior $\mu = \Pr(G \mid e)$. Write $\mu_1(\lambda) = \Pr(G \mid e = 1)$ and $\mu_0 = \Pr(G \mid s = 0) = \pi(1 - q)/(\pi(1 - q) + (1 - \pi)q)$. For $\lambda \in (0, 1]$, $\mu_0 = \Pr(G \mid e = 0)$; at $\lambda = 0$ a refusal has probability zero and μ_0 is the continuous extension.

The parameter λ summarizes what the components of Q^{I} secure for a single authority. A tenured faculty, an endowed seminary, a juristic school with its own revenue, a regulator with a reputation to protect (Carpenter, 2010), or an editor who cannot be dismissed for a story all raise λ.

Proposition 6.2 (Credibility and sorting)**.** *In the endorsement model:*

(a) $\mu_1(0) = \pi$*: a captured authority's endorsement carries no information.*

(b) $\mu_1(\lambda)$ *is strictly increasing in* λ*, and* $\mu_0 < \pi$ *does not depend on* λ*.*

(c) $\mathbb{E}[\mu \mid G]$ *is strictly increasing and* $\mathbb{E}[\mu \mid B]$ *is strictly decreasing in* λ*.*

Proof. Let $a = \Pr(s = 0) = \pi(1 - q) + (1 - \pi)q$. A refusal occurs only when the authority is independent and $s = 0$, so $\Pr(e = 0) = \lambda a$ and, for $\lambda > 0$, $\Pr(G \mid e = 0) = \Pr(G \mid s = 0) = \pi(1 - q)/a = \mu_0$, which does not depend on λ. Since $q > \frac{1}{2}$, $\mu_0 < \pi$.

(a) and (b). By Bayes' rule,

$$\mu_1(\lambda) = \frac{\pi\big(1 - \lambda(1 - q)\big)}{\pi\big(1 - \lambda(1 - q)\big) + (1 - \pi)(1 - \lambda q)} = \Big(1 + \tfrac{1-\pi}{\pi}\, r(\lambda)\Big)^{-1}, \qquad r(\lambda) = \frac{1 - \lambda q}{1 - \lambda(1 - q)}.$$

At $\lambda = 0$, $r = 1$ and $\mu_1 = \pi$. Differentiating, $r'(\lambda) = (1 - 2q)/(1 - \lambda(1 - q))^2 < 0$, so μ_1 is strictly increasing.

(c) Posteriors are a martingale, so $\mathbb{E}[\mu] = \pi$, and $\mathbb{E}[\mu\, \mathbf{1}_G] = \mathbb{E}[\mu \Pr(G \mid e)] = \mathbb{E}[\mu^2]$. Hence $\mathbb{E}[\mu \mid G] = \mathbb{E}[\mu^2]/\pi$ and $\mathbb{E}[\mu \mid B] = (\pi - \mathbb{E}[\mu^2])/(1 - \pi)$. It suffices to show that $\mathbb{E}[\mu^2]$ is strictly increasing in λ. From the martingale identity, $\Pr(e = 1)\mu_1 = \pi - \lambda a \mu_0$, so $\mu_1 = (\pi - \lambda a \mu_0)/(1 - \lambda a)$ and

$$\mathbb{E}[\mu^2] = \frac{(\pi - \lambda a \mu_0)^2}{1 - \lambda a} + \lambda a \mu_0^2.$$

Differentiating and substituting μ_1,

$$\frac{d\,\mathbb{E}[\mu^2]}{d\lambda} = a\mu_1(\mu_1 - 2\mu_0) + a\mu_0^2 = a\,(\mu_1 - \mu_0)^2.$$

By (b), $\mu_1(\lambda) \geq \pi > \mu_0$ for every λ, and $a > 0$, so the derivative is strictly positive. □

Part (a) is the formal version of an old observation: praise from a courtier who must praise is worthless. Part (c) carries the argument further. Independent authorities are an asset to a coalition whose rules are sound and a liability to one whose rules are not. The same institution that lends

credibility to a sound public-health rule exposes an unsound one. This is the precise sense in which healthcare, science and religious institutions can either check or enable capture: their independence is what makes their endorsement worth having, and it is also what makes them dangerous to a coalition that needs endorsement for an unsound rule. The next subsection turns the result into a comparative static for a governing coalition.

6.3 The control and credibility trade-off

A governing coalition values public confidence in its rules and also values control over authorities for its own sake, because control lets it direct authoritative outputs toward other ends. Independence is largely fixed in advance, by charters, endowments, tenure rules and appointment procedures, before the coalition knows whether the rules it will later need endorsed are sound. We therefore model the choice of λ as an act of institutional design taken before θ is realized and observed by the public. Because the choice precedes θ, it carries no information about θ, and the public's prior remains π.

Proposition 6.3 (Control and credibility)**.** *Before θ is realized, a coalition chooses a publicly observed independence $\lambda \in [\lambda_{\min}, 1]$, where $\lambda_{\min} \in [0, 1)$ is the independence that existing institutions guarantee whatever the coalition does. Its expected payoff is*

$$W(\lambda; \rho) = \pi\, \beta_G\, \mathbb{E}[\mu \mid G](\lambda) + (1 - \pi)\, \beta_B\, \mathbb{E}[\mu \mid B](\lambda) + \rho\,(1 - \lambda),$$

where $\beta_G, \beta_B \geq 0$ are the values the coalition places on public confidence when its rule is sound and unsound, and $\rho \geq 0$ is the value of control. Let $\Delta = \beta_G - \beta_B$ and $S(\lambda) = \mathbb{E}[\mu^2](\lambda)$. Then:

(a) $W(\lambda; \rho) = \Delta\, S(\lambda) + \beta_B \pi + \rho(1 - \lambda)$, and S is strictly increasing and strictly convex.

(b) If $\Delta < 0$, or if $\Delta = 0$ and $\rho > 0$, the unique choice is $\lambda = \lambda_{\min}$.

(c) If $\Delta > 0$, every maximizer lies in $\{\lambda_{\min}, 1\}$, and $\lambda = 1$ is the unique maximizer if and only if $\Delta(S(1) - S(\lambda_{\min})) > \rho\,(1 - \lambda_{\min})$.

(d) For fixed Δ, every maximizer at a higher ρ is weakly below every maximizer at a lower ρ.

Proof. (a) By the identities in the proof of Proposition 6.2, $\pi\mathbb{E}[\mu \mid G] = S$ and $(1-\pi)\mathbb{E}[\mu \mid B] = \pi - S$. Substituting gives the expression for W. The same proof gives $S'(\lambda) = a(\mu_1(\lambda) - \mu_0)^2 > 0$. Since μ_1 is strictly increasing and $\mu_1 > \mu_0$, S' is strictly increasing, so S is strictly convex.

(b) If $\Delta < 0$, ΔS is strictly decreasing and $\rho(1 - \lambda)$ is nonincreasing, so W is strictly decreasing. If $\Delta = 0$ and $\rho > 0$, W is strictly decreasing through its last term. In both cases $\lambda_{\min}$ is the unique maximizer.

(c) If $\Delta > 0$, W is the sum of a strictly convex function and a linear one, hence strictly convex on $[\lambda_{\min}, 1]$, and a strictly convex function on a compact interval attains its maximum only at endpoints. Comparing the endpoints, $W(1) > W(\lambda_{\min})$ if and only if $\Delta(S(1) - S(\lambda_{\min})) > \rho(1 - \lambda_{\min})$.

(d) Let $\rho < \rho'$, let λ maximize $W(\cdot; \rho)$ and λ' maximize $W(\cdot; \rho')$. Optimality gives $W(\lambda; \rho) \geq W(\lambda'; \rho)$ and $W(\lambda'; \rho') \geq W(\lambda; \rho')$. Adding the two inequalities, the terms in S cancel and what remains is $(\lambda' - \lambda)(\rho - \rho') \geq 0$. Since $\rho - \rho' < 0$, $\lambda' \leq \lambda$. □

The proposition is a conditional comparative static, and its key parameter is alignment. A coalition values independent authorities only when it gains more from public confidence in sound rules than in unsound ones ($\Delta > 0$), which is the case when its interests run with the public's: a government that wants a vaccine taken because it works, not because a contract depends on it. A coalition whose gains do not depend on soundness ($\Delta < 0$, or $\Delta = 0$ with any positive value of

control), such as one that profits from compliance with whatever rule it issues, pushes independence to the floor. The floor $\lambda_{\min}$, set by endowments, tenure, constitutional protection and transnational appointment, therefore protects the public exactly where alignment fails. Because S is convex, the choice is all or nothing: aligned coalitions with a small value of control grant full independence, and the rest grant the minimum. Intermediate independence in the record reflects floors and constraints, not interior optima.

Interim capture is different. After institutions are set and the public has formed a conjecture $\hat{\lambda}$, a coalition that has learned θ may be able to lower actual independence covertly, by pressuring editors or steering grants, without the public revising $\hat{\lambda}$.

Proposition 6.4 (Covert capture)**.** *Suppose the public's posterior after an endorsement is $\mu_1(\hat{\lambda})$ for a fixed conjecture $\hat{\lambda} > 0$, while actual independence is λ. Then for $\theta \in \{G, B\}$,*

$$\mathbb{E}[\mu \mid \theta] = \mu_1(\hat{\lambda}) - \lambda \Pr(s = 0 \mid \theta)(\mu_1(\hat{\lambda}) - \mu_0),$$

which is strictly decreasing in λ for both types. The gain from a unit reduction of λ is $q(\mu_1(\hat{\lambda}) - \mu_0)$ when the rule is unsound and $(1-q)(\mu_1(\hat{\lambda}) - \mu_0)$ when it is sound, so it is larger by the factor $q/(1-q) > 1$ for a coalition whose rule is unsound.

Proof. Given θ, a refusal occurs with probability $\lambda \Pr(s = 0 \mid \theta)$ and yields the posterior μ_0; otherwise the authority endorses and the public's posterior is $\mu_1(\hat{\lambda})$. Taking the expectation gives the formula. Since $\mu_1(\hat{\lambda}) \geq \pi > \mu_0$ by Proposition 6.2(b), and $\Pr(s = 0 \mid B) = q$ and $\Pr(s = 0 \mid G) = 1 - q$ are positive, the expression is strictly decreasing in λ with the stated slopes. □

Covert capture always raises expected confidence in the short run, because the public cannot discount what it does not see. Its value is greater for coalitions whose rules are unsound, since independent authorities would refuse them more often. If covert capture has a cost, coalitions with unsound rules will therefore pay it more often, and restrictions on independent authorities should follow adverse findings. Hypothesis HI7 in Section 7 tests that prediction. If instead the coalition chose λ openly after learning θ, the choice would itself signal θ and the analysis would become a signaling game, which this paper does not solve. Commercialized but party-guided media in China (Stockmann, 2013) and the simulated democratic forms of informational autocracies (Guriev and Treisman, 2019) fit the covert pattern: control is exercised while the appearance of independence is maintained.

The covert case points to the main channel of deception. After an endorsement the public's posterior is $\mu_1(\hat{\lambda})$, while the true probability that the rule is sound given an endorsement is $\mu_1(\lambda)$.

Corollary 6.5 (Deception gap)**.** *The public overestimates the soundness of an endorsed rule, $\mu_1(\hat{\lambda}) > \mu_1(\lambda)$, if and only if $\hat{\lambda} > \lambda$. The gap $\mu_1(\hat{\lambda}) - \mu_1(\lambda)$ is strictly increasing in $\hat{\lambda}$ and strictly decreasing in λ.*

Proof. Immediate from the strict monotonicity of μ_1 in Proposition 6.2(b). □

Capture misleads exactly when audiences overestimate independence. Disclosure rules, conflict-of-interest registries and the publication of appointment and funding records exist to bring $\hat{\lambda}$ into line with λ. They do not make authorities independent; they make their dependence visible, which removes the gap.

6.4 Correlated control and the discovery of error

Most polities have many legitimating authorities. Whether they detect and disclose an error depends on how many of them are independent, and on whether their independence is correlated. Integration across pillars, in the source's sense, typically works by correlation: when one coalition appoints the editors, the rectors and the chief jurists, they are captured together or not at all.

Proposition 6.6 (Disclosure under correlated control)**.** *Let $K \geq 2$ authorities each be independent with probability $\lambda \in (0,1)$, with an arbitrary joint distribution of independence across authorities, and let N be the number of independent authorities. Given the set of independent authorities, each independently discovers and discloses a given error with probability $p \in (0,1)$; captured authorities never disclose. Let $D = \Pr(\textit{at least one disclosure})$.*

(a) For every joint distribution, $\lambda(1-(1-p)^K) \leq D \leq 1-(1-p)^{\lambda K}$.

(b) The lower bound is attained if and only if independence is all or nothing, $N \in \{0, K\}$ almost surely, which is the case of perfectly correlated control.

(c) If independence is drawn independently across authorities, $D = 1-(1-\lambda p)^K$, which lies strictly above the lower bound.

Proof. Conditional on N, no disclosure occurs with probability $(1-p)^N$, so $D = 1 - \mathbb{E}[g(N)]$ with $g(x) = (1-p)^x$, a strictly convex function. By linearity of expectation $\mathbb{E}[N] = \lambda K$.

(a) For the upper bound, Jensen's inequality gives $\mathbb{E}[g(N)] \geq g(\mathbb{E}N) = (1-p)^{\lambda K}$. For the lower bound, strict convexity places g below its chord on $[0, K]$: $g(x) \leq (1-x/K)g(0) + (x/K)g(K)$, with equality only at $x \in \{0, K\}$. Taking expectations, $\mathbb{E}[g(N)] \leq (1-\lambda) + \lambda(1-p)^K$, so $D \geq \lambda(1-(1-p)^K)$.

(b) The chord inequality is strict for $x \in (0, K)$, so equality in expectation requires $\Pr(N \in \{0, K\}) = 1$. Conversely, if $N \in \{0, K\}$ almost surely then $\Pr(N = K) = \lambda$ and $D = \lambda(1-(1-p)^K)$.

(c) If independence is independent across authorities, N is binomial with parameters K and λ, and $\mathbb{E}[(1-p)^N] = (1-\lambda+\lambda(1-p))^K = (1-\lambda p)^K$. Since $K \geq 2$ and $\lambda \in (0,1)$, $\Pr(0 < N < K) > 0$, so by (b) the bound is strict. □

Corollary 6.7 (Integration lowers discovery)**.** *Holding fixed the number of authorities and the average probability that each is independent, a change in appointment structure that makes the independence of authorities perfectly correlated lowers the probability that an error is disclosed from $1-(1-\lambda p)^K$ to $\lambda(1-(1-p)^K)$.*

The corollary separates two things that are often merged. Concentration H^{I} concerns how many authorities there are and how control is shared among them. Integration concerns whether control over them moves together, which is what high C_{PI} or C_{FI} produces when one coalition appoints or funds all of them. A polity can have many universities, newspapers and juristic schools, and so low measured concentration, and still have low discovery if one coalition appoints their heads. The upper bound in (a) is attained only when λK is an integer and $N = \lambda K$ almost surely; otherwise the maximum is reached by a distribution on $\lfloor \lambda K \rfloor$ and $\lceil \lambda K \rceil$. The bound adds a design point: discovery is highest when the number of independent authorities is least variable, which plural appointment by rival bodies secures better than independent chance. This refines source hypothesis H4 on epistemic checks and source Proposition 2 on the interruption of transmission (Largo, 2026).

6.5 Fragility of displayed legitimacy

Ideological control often aims at public displays: attendance, applause, signatures, votes in uncontested elections. The beliefs those displays reveal are a separate matter. A threshold model, of the kind Granovetter (1978) introduced for collective behavior and Kuran (1991) applied to preference falsification, shows how far the two can diverge.

Proposition 6.8 (Fragility)**.** *Let n members have thresholds $\tau_i \in [0,1]$: member i expresses dissent publicly when the share of members already doing so is at least τ_i. Let $\Phi(x) = \frac{1}{n}\#\{i : \tau_i \le x\}$ and $x_0 = 0$, $x_{k+1} = \Phi(x_k)$.*

(a) The sequence (x_k) is nondecreasing and reaches, within $n+1$ steps, the smallest fixed point x^* of Φ.

(b) There are threshold profiles τ and τ' that differ only in one member's threshold, by $1/n$, with $x^*(\tau) = 1$ and $x^*(\tau') = 0$.

Proof. (a) Φ is nondecreasing. Since $x_1 = \Phi(0) \ge 0 = x_0$, induction gives $x_{k+1} = \Phi(x_k) \ge \Phi(x_{k-1}) = x_k$. The sequence takes values in $\{0, 1/n, \dots, 1\}$, so it is constant from some step $k \le n$ on, at a fixed point $\bar{x}$. If y is any fixed point, then $x_0 \le y$, and $x_k \le y$ implies $x_{k+1} = \Phi(x_k) \le \Phi(y) = y$, so $\bar{x} \le y$.

(b) Let $\tau_i = (i-1)/n$ for $i = 1, \dots, n$. Then $\Phi(k/n) = (k+1)/n$ for $k < n$, so $x_k = k/n$ and $x^* = 1$. Let τ' agree with τ except $\tau'_1 = 1/n$. No member has threshold 0, so $\Phi'(0) = 0$ and $x^*(\tau') = 0$. □

Two populations whose private dispositions are almost identical can display opposite public states. Displayed support is therefore not a sufficient statistic for private belief, and the gap is largest where expression is costly, which is where Q^{I} is low. The result explains why regimes that control the ideological layer can appear stable and then collapse quickly, as Kuran (1991) argued for 1989. It also constrains measurement: in polities with low contestability, attendance, applause and uncontested votes cannot validate claims about normative legitimacy.

6.6 Coupling of belief to coercion and production

The series assigns two of the fifteen cross-pillar pairs to this Part for primary treatment.

The coupling C_{IC} runs in both directions. Control of belief needs enforcement when it goes beyond persuasion: heresy and blasphemy prosecutions, censorship offices, the licensing of presses and the dismissal of dissenting teachers all require a sanctioning apparatus. The Spanish Inquisition joined doctrinal judgment to coercive power (Kamen, 1997), and the censorship offices of party-states worked with the security services. Coercion in turn needs belief. Soldiers and police carry out orders when they expect others to do so, and Proposition 6.8 shows how quickly that expectation can fail. Greitens (2016) and the other sources of Part V analyze the coercive institutions themselves, and Part V shows that the refusal of manifestly unlawful orders, one of its checks on coercion, depends on the legitimating beliefs studied here. For the ideological layer, C_{IC} is the extent to which one coalition both decides what may be said and commands the organizations that punish saying otherwise. It is high where religious courts or security services enforce doctrine and low where independent courts protect expression.

The coupling C_{IW} concerns the beliefs of the producing population and the institutions that shape them. Belief-producing specialists need a surplus to live on, a dependency of the ideological on the productive pillar that Part VII derives in general. In the other direction, beliefs shape work

and compliance. Levi (1988) showed that taxpayers' compliance is quasi-voluntary: it depends on their belief that others comply and that rulers keep their bargains. Becker and Woessmann (2009) found that Protestant regions of Prussia had higher literacy, which they attributed to the religious emphasis on reading scripture, and that literacy rather than a work ethic accounts for much of their economic advantage. Gellner (1983) argued that industrial economies require a standardized literate culture that only a mass education system can supply, which makes control of schooling central to modern states. Bourdieu and Passeron (1977) argued that schooling also reproduces class position under the appearance of merit. The coupling is high where one coalition controls both the conditions of work and the institutions that teach workers what to expect of it, and low where producers have authorities of their own, such as the labor press and educational associations of social democracies. Part VI treats everyday resistance (Scott, 1985), which shows that producers do not simply absorb dominant beliefs, and its latent-veto result, that producers can threaten a system's persistence by withdrawing the production it needs (Proposition 3.6 of Part VI), explains why rulers need producers to believe that compliance is expected of everyone.

6.7 Checks

Each component of Q^{I} maps onto a parameter of the models above. Independence raises λ and, through the institutional floor $\lambda_{\min}$, protects the public when the coalition's interests are not aligned with soundness (Proposition 6.3) and when it would capture authorities covertly (Proposition 6.4). Disclosure aligns the public's estimate $\hat{\lambda}$ with the truth and closes the deception gap (Corollary 6.5). Plurality with separate appointment raises the number of authorities and lowers the correlation of their capture (Proposition 6.6). Reviewability raises the discovery probability p. Exit lets audiences abandon authorities that lose credibility, which gives authorities a reputational reason to remain independent, the mechanism behind both the demand-side account of Gentzkow and Shapiro (2006) and the regulator reputation of Carpenter (2010).

The historical record shows each check operating. Scholars refused the Abbasid caliph's doctrinal demand (Crone and Hinds, 1986). Endowments gave madrasas a revenue base of their own (Makdisi, 1981). The transnational Church limited the claims of crowns (Grzymala-Busse, 2023), and Jansenist and Gallican counter-publics eroded royal sacrality (Van Kley, 1996). Competition among outlets raises the cost of media capture (Besley and Prat, 2006), and meta-research exposed sponsorship bias in clinical research (Lundh et al., 2017). These are the forms that source Proposition 2, that checks can interrupt the transmission of concentrated power, takes in the ideological layer (Largo, 2026).

7 Measurement

The constructs of this Part are measurable from documents. Table 3 lists indicators for each. All are subject to the source's requirements: weights and thresholds are preregistered, the population of events is defined before data collection rather than drawn from scandals, and measurement invariance is tested before indicators are compared across regime types (Largo, 2026).

The hypotheses below refine the source's hypotheses for this pillar. Each has a stated falsifier. Several are drawn directly from the propositions of Section 6; the others restate claims from the literature in testable form.

Three identification problems deserve emphasis. Discovery and capture are confounded: a high rate of documented error may mean many errors or strong review, so HI4 must compare correction given comparable exposure, as source hypothesis H4 already notes. Selection of cases from scandals biases every estimate toward finding capture, which is why the population must be preregistered.

Table 3: Indicators for the ideological layer. Each construct can be coded from public documents or existing datasets.

Construct	Indicator	Data or design
Concentration H^{I}	Shares of appointments, accreditation decisions and audience reach held by each organization	Ownership data (Djankov et al., 2003) and appointment records
Independence	Tenure protection, editorial autonomy and academic and juristic freedom	Expert-coded indicators (Coppedge et al., 2011)
Disclosure	Share of trials, articles and broadcasts with published funding and conflict statements	Trial registries and journal policies
Reviewability	Rates of correction, retraction and successful replication	Retraction and replication records
Exit	Legal penalties for unlicensed publication, apostasy or blasphemy	Religion and state coding (Fox, 2008)
Censorship practice	What is removed and what is left standing	Audit and observational designs (King et al., 2013)
Correlated control	Share of authority heads appointed by the same body	Appointment registers and party lists
Coupling C_{IC}	Prosecutions and sanctions for expression and the bodies that order them	Court and security records
Platform reach	Shares of ranked and moderated reach held by each platform	Platform disclosures and feed experiments (Guess et al., 2023)

And indicators of displayed support are invalid where contestability is low, by Proposition 6.8, so private measures such as list experiments are needed. None of these problems is peculiar to any regime type, and each applies with equal force to all six.

8 A typological reading of Revelation 13

8.1 Method

The measurement program above states what evidence would show capture of the ideological pillar. The series pairs it with a normative image of capture drawn from Revelation 13, read typologically. On this reading the beast is a recurring pattern of power that claims ultimate loyalty, and Babylon is the city that organizes commerce and worship around that claim in any age. Bauckham (1993) read Rome as the paradigm of such power, and Beale (1999) defended an idealist reading that treats the images as transhistorical. The contextual reading anchors the images in the Roman province of Asia, its imperial cult and its trade (Koester, 2014; Friesen, 2001; Kraybill, 1996). Futurist readings, which take the text to predict a specific future order or person, are excluded by the series' guardrails. Their reception history shows why: Fuller (1995) and Boyer (1992) document repeated identifications of the beast with contemporary figures, each of which failed. Historicist readings that identified the papacy with the beast belonged to the polemics of the Reformation, as the reception histories of Emmerson and McGinn (1992), McGinn (1998) and Kovacs and Rowland (2004) document. Any identification of a church, a faith, a profession or a regime with the beast is excluded here for the same reason. What follows is an interpretive claim of this project, offered as

Table 4: Hypotheses for the ideological pillar, with the observation that would count against each.

Hypothesis	Prediction	Falsifier
HI1 State media ownership	Greater state ownership of media is associated with weaker accountability outcomes (Djankov et al., 2003).	No association after controls, or public broadcasters with plural governance behaving like state-owned media.
HI2 Demand and supply of slant	Slant tracks audience composition more than owner identity (Gentzkow and Shapiro, 2006).	Changes of owner that shift content with audience composition held constant.
HI3 Sponsorship and disclosure	Disclosure and registration rules narrow the gap between sponsored and unsponsored conclusions (Lundh et al., 2017).	A gap that does not narrow after such rules are introduced.
HI4 Correlated control	With average independence fixed, systems where one body appoints most authorities show less documented error correction.	Equal or higher correction under common appointment.
HI5 Fragmentation of authority	Mass education and new media raise the number of recognized religious and scholarly authorities (Eickelman and Piscatori, 1996; Eisenstein, 1979).	No rise in recognized claimants after such expansions.
HI6 Credibility premium	Outlets perceived as independent persuade more than official outlets (Stockmann, 2013; Enikolopov et al., 2011).	No difference in persuasive effect.
HI7 Attacks follow adverse findings	Restrictions on independent authorities follow findings unfavorable to the governing coalition (Proposition 6.4).	Restrictions unrelated in timing to prior adverse findings.
HI8 Hidden opinion	The gap between displayed and privately reported support is larger where contestability is lower (Kuran, 1991).	No relation between contestability and the gap.

an analogy that aids comparison, not as a settled exegetical result.

8.2 The second beast as legitimating authority

Revelation 13 presents two beasts. The first, from the sea, holds authority and makes war. The second rises from the land and does not rule. In the King James text it "had two horns like a lamb, and he spake as a dragon" (13:11). It "exerciseth all the power of the first beast before him, and causeth the earth and them which dwell therein to worship the first beast" (13:12), and it "deceiveth them that dwell on the earth by the means of those miracles which he had power to do" (13:14). Later chapters name it "the false prophet that wrought miracles before him" (19:20).

The second beast performs the function this paper studies. It produces belief in the authority of the first beast, and it does so through public signs performed "in the sight of men" (13:13), which are public events in the sense of Definition 2.4. The contextual reading identifies it with the agents of the imperial cult in Asia. Price (1984) showed that the cult in Asia Minor was organized largely by the cities and their elites, who used it to represent imperial power within their own civic religion, and Friesen (2001) examined its provincial institutions. The legitimation of Roman rule was thus supplied locally, by elites who gained standing from it. In the vocabulary of Section 6, it rested on privileged access exchanged for endorsement.

The mismatch between the lamb's horns and the dragon's voice is the typological image of the

deception gap in Corollary 6.5. An authority that appears independent and gentle, so that its audience assigns it high $\hat{\lambda}$, speaks entirely for the ruler, so that its true λ is zero. The text calls the result deception. The model calls it a posterior that exceeds the truth by an amount that grows with the gap between perceived and actual independence.

8.3 Image, enforcement and exchange

The second beast makes an image that speaks and causes "that as many as would not worship the image of the beast should be killed" (13:15). The image joins legitimation to coercion, which is the coupling C_{IC} at its maximum. The text then joins both to economic participation: the second beast causes all "to receive a mark in their right hand, or in their foreheads: And that no man might buy or sell, save he that had the mark" (13:16 to 13:17). Displayed allegiance becomes the condition of exchange, which couples the ideological pillar to the financial and productive pillars that Parts I and VI analyze. Kraybill (1996) argued that for Christians in Asia, participation in trade guilds and commerce carried obligations to honor the imperial cult, so the image had a concrete social referent. In Revelation 18 the merchants of Babylon "were the great men of the earth; for by thy sorceries were all nations deceived" (18:23), an image of commerce and deception bound together that Bauckham (1993) reads as part of an economic critique of Rome.

Proposition 6.8 supplies a reading of compelled worship. Worship enforced on pain of death is a display whose relation to private belief is unknown, and the text treats it as such: the faithful in Revelation are those who refuse the display. Collins (1984) read the book as a response to a crisis perceived by a minority community, and its images express that community's experience of a public culture in which displayed loyalty was demanded.

8.4 Scope of the reading

The normative warning in the text attaches to Proposition B, not to Proposition A. Revelation does not condemn teaching, worship, scholarship or the existence of authorities that shape belief; it is itself an act of witness that seeks to shape belief. Its target is a configuration in which legitimating authority is fused with rule and coercion, independence is zero, and exit from the displayed allegiance is closed. In the terms of this paper, that is $\lambda = 0$, Q^{I} near zero, and high C_{PI}, C_{IC} and coupling to exchange. Ellul (1977) read the beasts as political power and its propaganda, and Wink (1992) described the powers as institutions whose spirituality can become a domination system. Both readings are compatible with the structural interpretation adopted here, on which any authority, religious or secular, can occupy the position of the second beast when its speech is captured by rule. No church, university, newsroom or medical body occupies that position by virtue of what it is.

9 Objections and scope conditions

9.1 Vacuity

If the legitimation function were defined so broadly that any social activity belonged to it, the necessity claim would be empty. The definition excludes much. A practice belongs to the class only if it regularly changes members' beliefs about rules, officials or relevant facts and if members rely on the changed beliefs to coordinate. Private reflection, idiosyncratic opinion and persuasion without effect on compliance are excluded. The theorem can fail if its assumptions fail, and Section 3 names the observation that would refute it. A claim with a stated falsifier is not vacuous.

9.2 Ideology critique

A Marxian critic may object that the paper treats legitimation too neutrally, as coordination, when its historical role has been to secure acceptance of domination. The endorsement model answers part of the objection. It identifies when legitimation misleads, through captured endorsement and the deception gap, and it predicts that coalitions whose gains do not depend on soundness push independence to its floor (Proposition 6.3(b)), and that covert capture is worth more to coalitions whose rules are unsound (Proposition 6.4). What the model does not do is presume that every legitimating belief is false. That presumption would require the counterfactual preferences that Lukes (2005) acknowledged are hard to observe, and it would make the framework unable to distinguish a sound public-health rule from an unsound one. The difference between the two is the substance of Proposition 6.2.

9.3 Expertise and faith

A defender of science, medicine or religion may object that classing these institutions as ideological power disparages them. The classification concerns the function they perform for coordination, not the truth of what they teach. A physician who certifies that a vaccine is safe, a jurist who explains a rule of inheritance and a priest who instructs a congregation each change beliefs that others rely on. The results of this paper give independent authorities their value: their endorsement is informative because it is independent, and they protect the public precisely when a coalition's rule is unsound. The source framework's caution applies throughout: a hospital is not a propaganda organ and a professor is not an agent of command (Largo, 2026). The paper is silent on the truth of any faith's claims, and the theological reading of Section 8 rests on a distinction that Revelation itself draws between witness and enforced display.

9.4 Measurement of belief

Belief is internal, while the control share s^{I} is defined over observable outputs such as appointments, accreditation and reach. A critic may object that control over outputs is not control over minds. The objection is correct and is the reason the definition is restricted to outputs. Persuasion studies (DellaVigna and Kaplan, 2007; Enikolopov et al., 2011) show that control over channels changes behavior, which is the evidence that edges in this layer are real. Whether a given share of outputs translates into a given share of influence is an empirical question that the indicators of Table 3 are designed to address.

9.5 Scope conditions

The comparative claims are qualitative codings from secondary literature. Nothing in Section 5 estimates the frequency of capture in any regime type, and the historical illustrations are not a sample. They show how the function is held in each type; they do not show how often it is captured.

The regime types are ideal types with large internal variation. The comparative sections record that variation for Islamic states and Catholic monarchies in particular, and it holds for the other four types as well.

The formal models are deliberately simple. They treat soundness as binary, independence as a probability and disclosure as independent across authorities given their type. Richer models would change magnitudes. The directions established by the proofs depend only on the monotonicity and convexity properties used there.

10 Conclusion

Every persistent human collective with turnover and locally variable rules implements a recognition function: a recurring public practice that makes its rules common knowledge. That much follows from the coordination axiom, and the stateless societies of the ethnographic record show the function carried by talk, ridicule, ritual and kinship idiom without any office. Belief in the rightness of rules does not follow, and regimes vary widely in how much of it they command.

The question that separates regimes is therefore not whether they have an ideological pillar but how independent its authorities are. A captured authority's endorsement carries no information. Independence helps coalitions with sound rules and hurts coalitions with unsound ones, so the institutional floor of independence, secured by endowments, tenure, constitutional protection or transnational appointment, protects the public in the cases where protection matters. Coordinated control of many authorities lowers the chance that an error is disclosed even when their average independence is unchanged, which is why integration across pillars, and not concentration within one, is the condition to measure.

The six regime types distribute the function differently. Market economies couple it to finance, party-states to appointment and coercion, social democracies to labor organizations, Islamic states and Catholic monarchies to religious authorities whose independence from rulers varied from substantial to nil, and secular democracies to a legally plural public sphere. The comparative codings remain qualitative, and the hypotheses of Section 7 state what evidence would overturn them. The existence of the pillar is universal, as Proposition A holds; its capture is a contingent and measurable condition, as Proposition B holds, and the first does not entail the second.

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